Breaking the Spurious Causality of Conditional Generation via Fairness Intervention with Corrective Sampling

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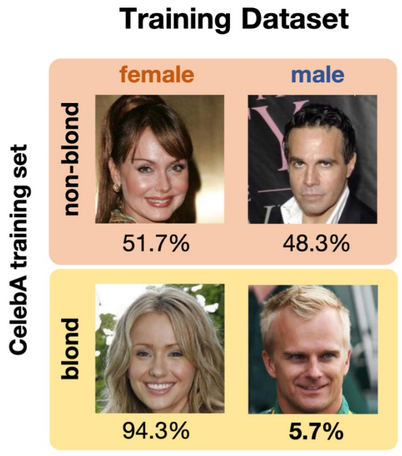
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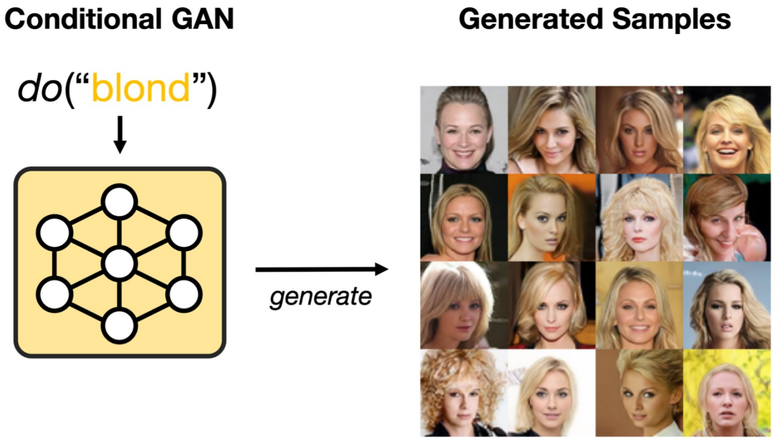
TL;DR: Spurious correlations in datasets can lead to spurious causal misconceptions in conditional generative models – and how to fix this

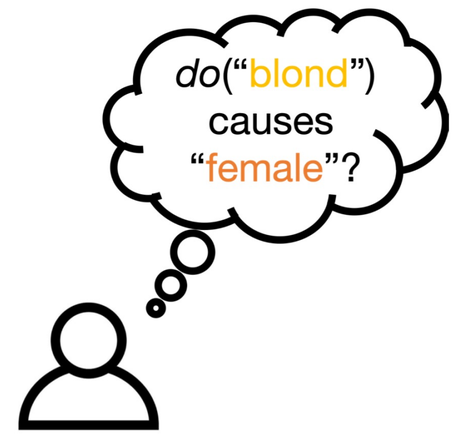
Motivation: cGANs can lead to spurious causal misconceptions



Spurious correlation in datasets (blond males are minority groups)

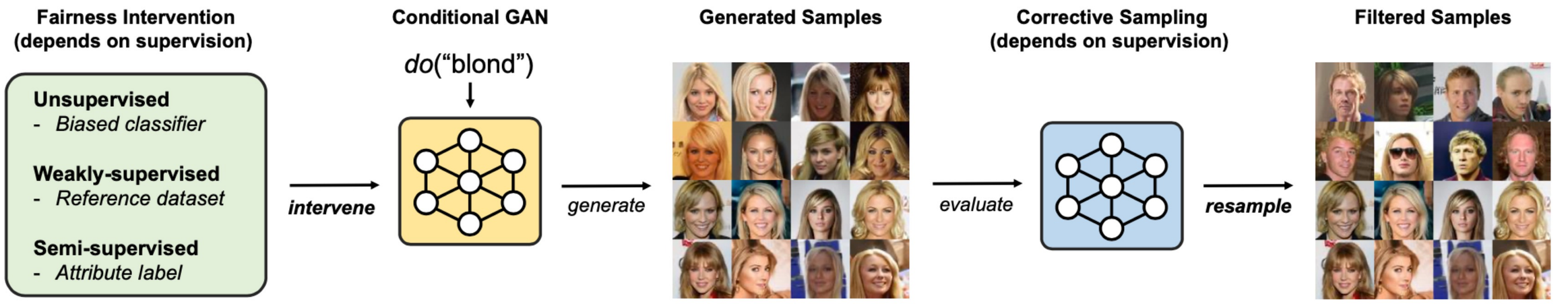
When used for training cGANs… the *spurious correlation* can lead to the *spurious causal misconception* a.k.a. **spurious causality** of cGANs





In cGANs, user **acts** a condition (e.g., generate "blond" images), which can lead to the **biased outcomes** (e.g., only generate "blond females")
→ The user may have a **misconception** that the condition y and the outcome attribute a has a **casual relation** $\mathrm{do}(y) \to a$

Method: Fairness Intervention w/ Corrective Sampling (FICS)



FICS: 2-step approach to fix your cGAN

- **Fairness Intervention (FI)**: Intervene the generator to synthesize *more minority samples* during the training phase (upweighted training)
- **Corrective Sampling (CS)**: Correct the generator to follow the *desired attribute distribution* during the sampling phase (rejection sampling)

How to intervene – depends on the level of fairness supervisions

- **Unsupervised**: Guide the model to follow the attribute ratio of the *data distribution* $p_{\mathrm{data}}(a|y)$
  - This can be done by our **novel observation** that classifiers and cGANs suffer from the **similar** spurious correlations
  - Based on this observation, we utilize a biased classifier $p(y|x)$ to promote a cGAN $p(x|y)$ to synthesize more minority samples
- **Supervised**: Guide the model follow the attribute ratio of the desired *fair distribution* $p_{\mathrm{fair}}(a|y)$
  - Weakly-supervised: Desired fairness is specified by a reference set $\mathcal{D}_{\mathrm{val}}$
  - Semi-supervised: Exact attribute labels a are annotated for a few samples

Experiments: Balance the generation of minor groups

Unsupervised setup

- Follow the original attribute ratio (less discrepancy)

	Intra-FID (↓)		Minor in Blond (%)		
	Blond	Non-blond	Male	Eyeglasses	No lipstick
Dataset			5.72	1.66	18.78
Estimated			5.85	1.85	17.61
Unsupervised					
cGAN (Mirza & Osindero, 2014)	1.86	1.64	3.97	0.88	11.04
Dia-GAN (Lee et al., 2021)	1.88	1.84	5.06	1.49	14.10
FICS (ours)	**1.78**	**1.38**	**6.13**	**1.90**	**16.34**

Supervised setup

- Follow the desired target attribute ratio (specified by user)

	Target Male ratio: 30%		Target Male ratio: 50%	
	Fair Intra-FID (↓) Blond	Occurrence of Male in Blond (%)	Fair Intra-FID (↓) Blond	Occurrence of Male in Blond (%)
Reference		30.00		50.00
Estimated		28.60		47.15
Weakly-supervised				
Choi et al. (2020)	10.71	7.60	10.70	18.34
FICS (ours)	**10.36**	**15.65**	**10.24**	**27.04**
Semi-supervised				
FairGen (Tan et al., 2020)	12.71	**29.54**	11.98	49.76
FICS (ours)	**10.56**	25.72	**11.68**	**47.11**



(a) Vanilla BigGAN (Brock et al., 2019) (b) FICS (ours)

Check our TMLR paper for further details!

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Paper Talk